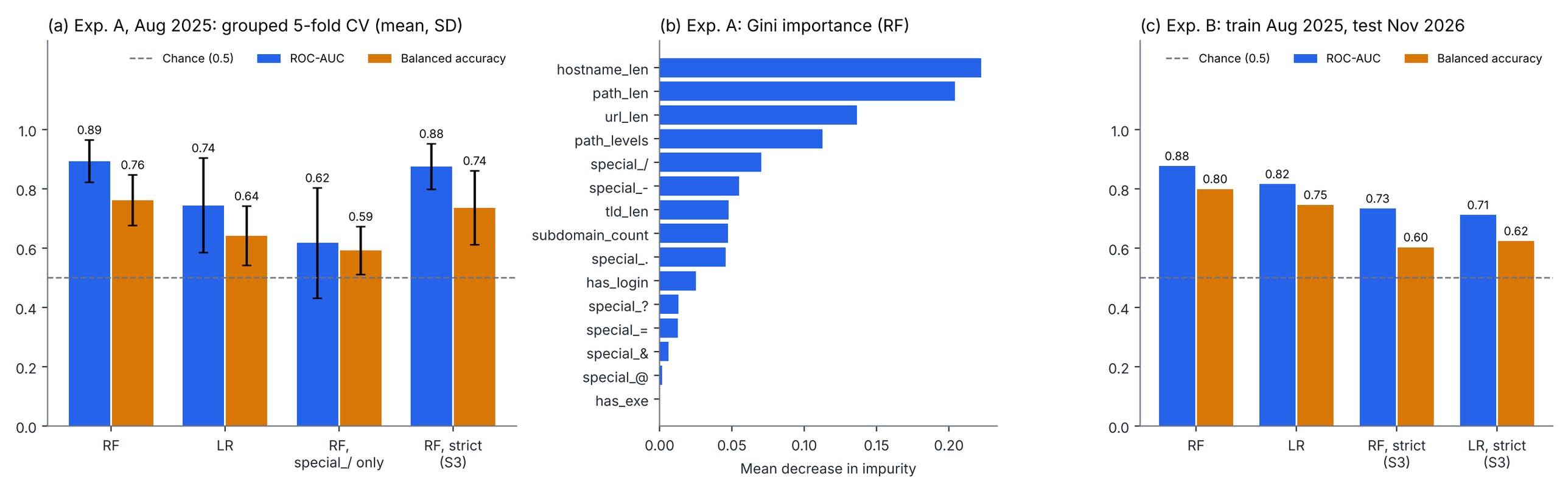
(a) Exp. A, Aug 2025: grouped 5-fold CV (mean, SD)
Chance (0.5)
ROC-AUC
Balanced accuracy
1.0
0.8
0.6
0.4
0.2
0.0
0.89
0.76
0.74
0.64
0.62
0.59
0.88
0.74
RF
LR
RF, special_/ only
RF, strict (S3)
(b) Exp. A: Gini importance (RF)
hostname_len
path_len
url_len
path_levels
special_/
special_-
tld_len
subdomain_count
special_.
has_login
special_?
special_=
special_&
special_@
has_exe
0.00
0.05
0.10
0.15
0.20
Mean decrease in impurity
(c) Exp. B: train Aug 2025, test Nov 2026
Chance (0.5)
ROC-AUC
Balanced accuracy
1.0
0.8
0.6
0.4
0.2
0.0
0.88
0.80
0.82
0.75
0.73
0.60
0.71
0.62
RF
LR
RF, strict (S3)
LR, strict (S3)